

User Guide

Snap Button Defect Detection System

Getting started

System Overview

What Does This System Do?

This automated button defect detection system uses ML and computer vision to inspect buttons on garments for various quality defects, including:

- ✓ **Button Count Verification** - Ensures the correct number of buttons
- ✓ **Distance Measurement** - Checks spacing between buttons
- ✓ **YOLO Based Defect Detection** - Identifies physical defects (bend, machine damage)
- ✓ **Color Matching** - Verifies button color matches fabric
- ✓ **Automatic Documentation** - Logs all detections to a CSV file

Key Features

-  **Real-time Camera Monitoring** - Live video feed
-  **Motion-triggered Capture** - Automatic detection when the garment passes
-  **Multiple Style Profiles** - Save different button configurations
-  **Image Archiving** - Saves original and annotated images
-  **CSV Logging** - Detailed records for quality control
-  **Touch Screen Interface** - Easy to use on a 7-inch display

System Requirements

- **Hardware:**
 - 7-inch touchscreen display (800×480)
 - USB camera
 - Mini PC with Python 3.10
 - ESP32 WiFi module (built-in)
 - Optional: Arduino/PLC connected to COM3 for defect signaling
- **Software:**
 - Python 3.10 or higher
 - Required packages: PySide6, OpenCV, YOLO, NumPy

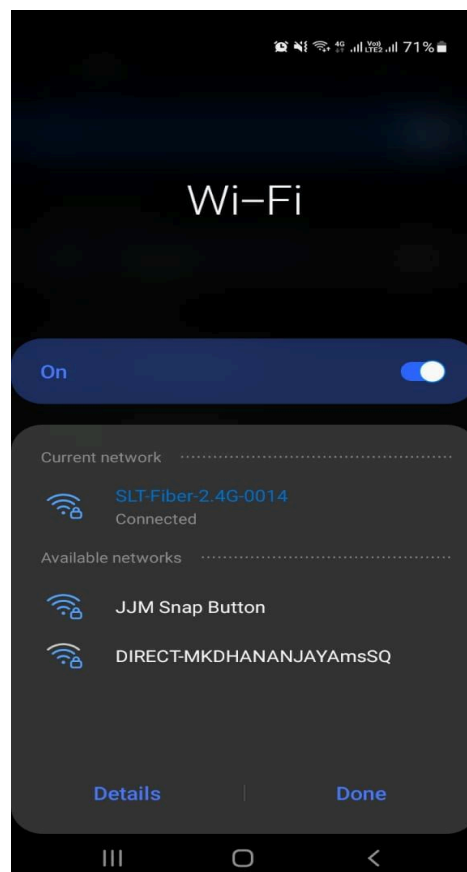
WiFi Setup (First Time Only)

The system includes an ESP32-based WiFi module for network connectivity.

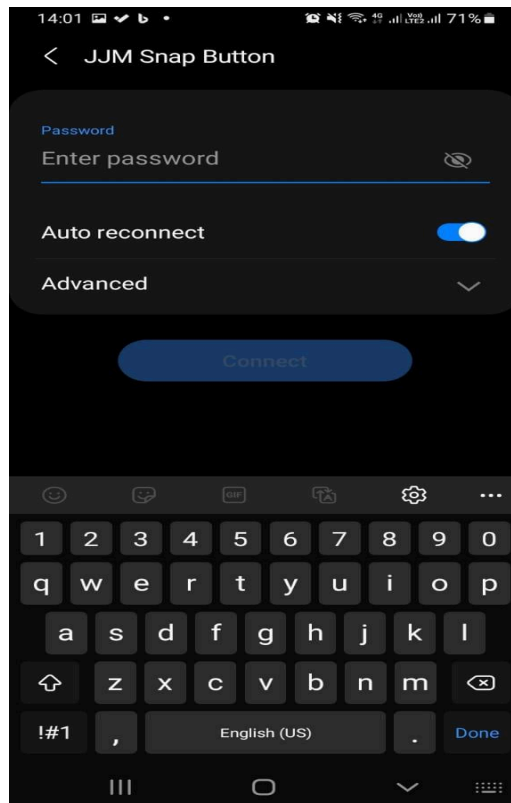
Step 1: Connect to Access Point

When the system starts for the first time (or if not connected to WiFi):

1. **On your phone, tablet, or laptop:**
 - Open WiFi settings
 - Look for a network named **"JJM Snap Button."**
 - Connect to the **"JJM Snap Button"** access point

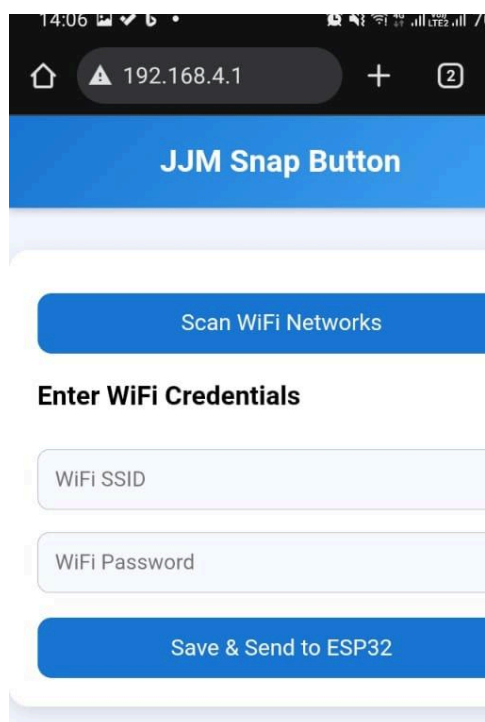


- The password “**12345678**” is required for the access point



Step 2: Open Configuration Dashboard

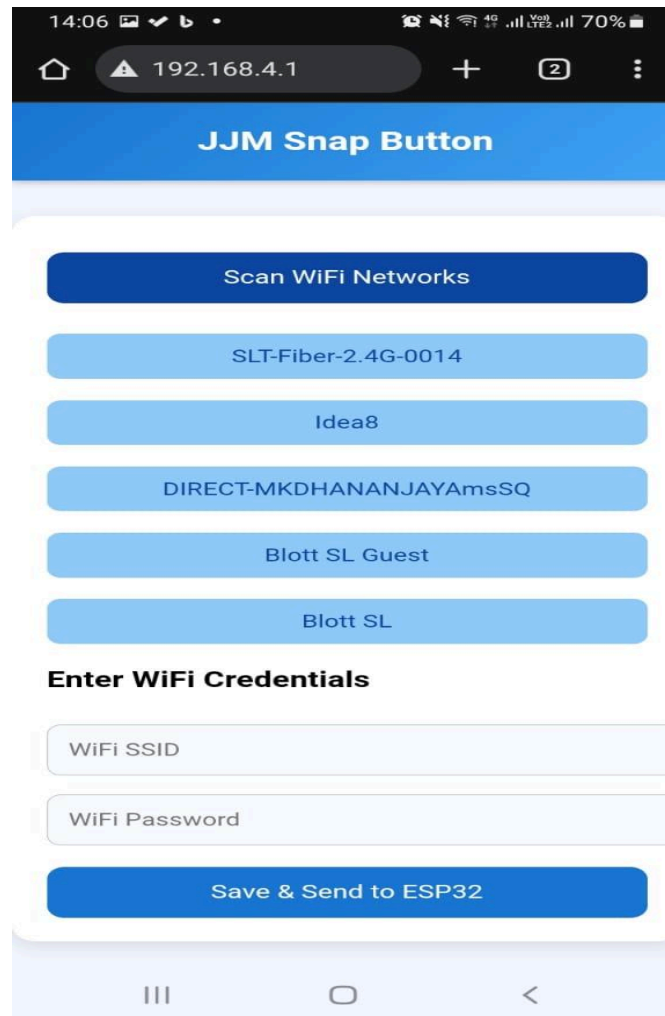
2. **Open a web browser** (Chrome, Firefox, Safari, etc.)
 - Go to: <http://192.168.4.1>
 - The WiFi configuration dashboard will load



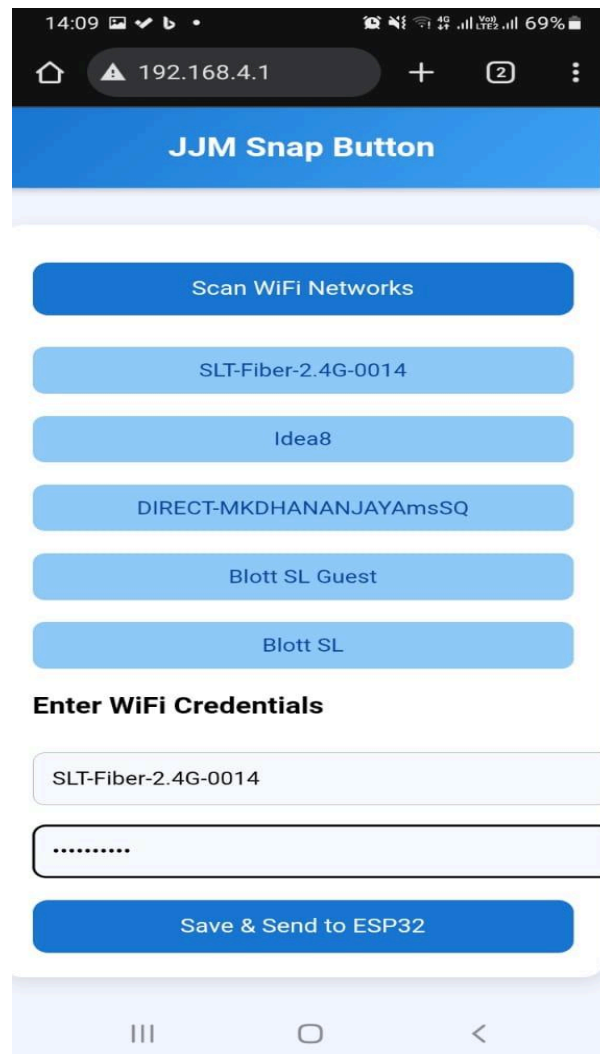
Step 3: Configure WiFi Connection

3. On the dashboard:

- Click "**Scan WiFi Networks**," a button



- Wait for the list of available WiFi networks to appear
- **Select your WiFi network** from the list
- **Enter the WiFi password** in the password field
- Click "**Save**"



4. Wait for confirmation:

- The dashboard will show "Connecting..."
- After 10-30 seconds, you'll see "Connected successfully!"
- The ESP32 will remember this network

Step 4: Verify Connection

5. Disconnect from the "JJM Snap Button" access point:

- Return to your device's WiFi settings
- Connect back to your normal WiFi network
- The detection system is now connected to your WiFi

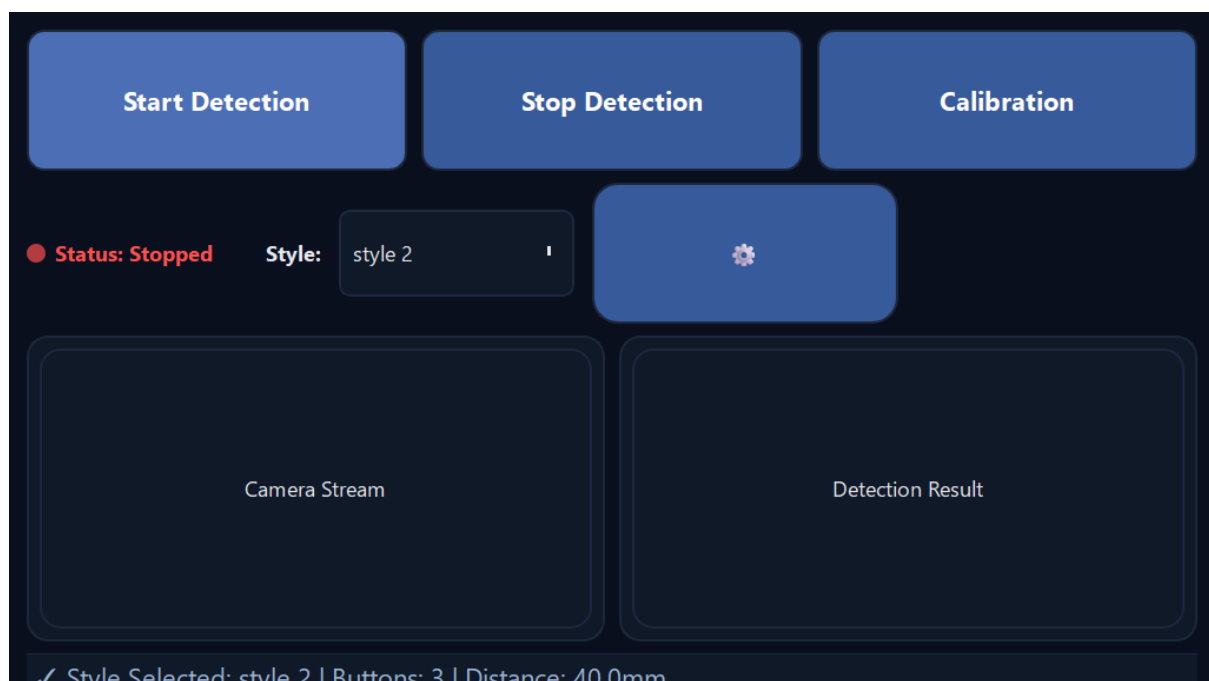
✅ **Auto-Connect Feature:** Once configured, the system will **automatically connect** to your saved WiFi network every time it starts. You only need to configure it once!

⚠ Important Notes:

- The "JJM Snap Button" access point only appears when the system cannot connect to the saved WiFi
- If you change your WiFi password, repeat the setup process
- If you move to a different location, you'll need to reconfigure the WiFi

First Time Application Launch

1. **Ensure WiFi is connected** (see above)
2. Connect the camera to your computer
3. Connect the touchscreen display
4. The system will open in **full-screen mode**



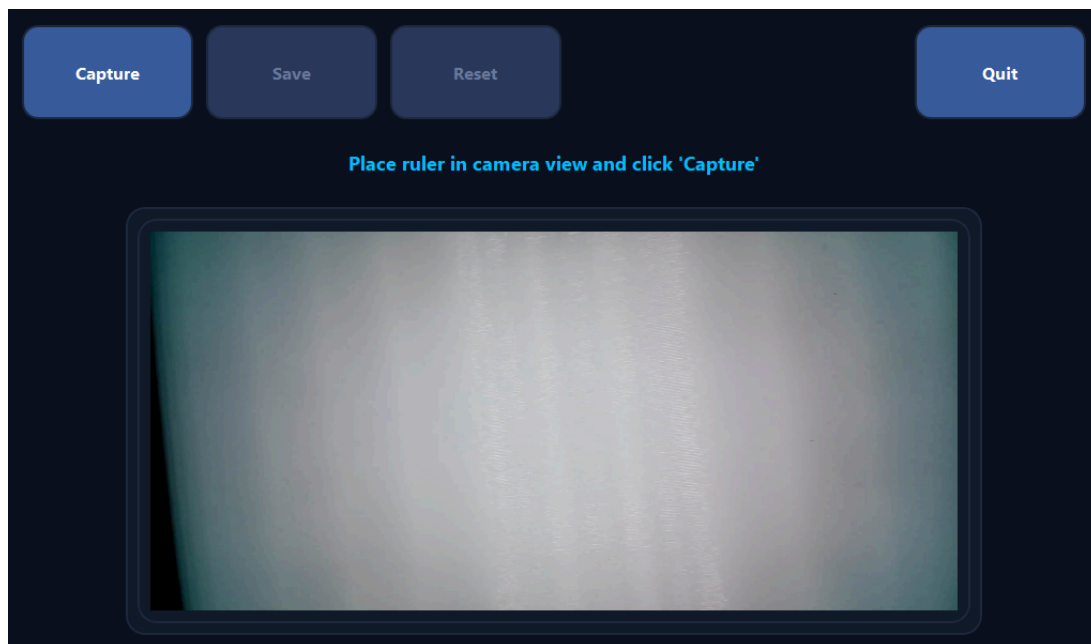
Initial Setup

Step 1: Camera Calibration

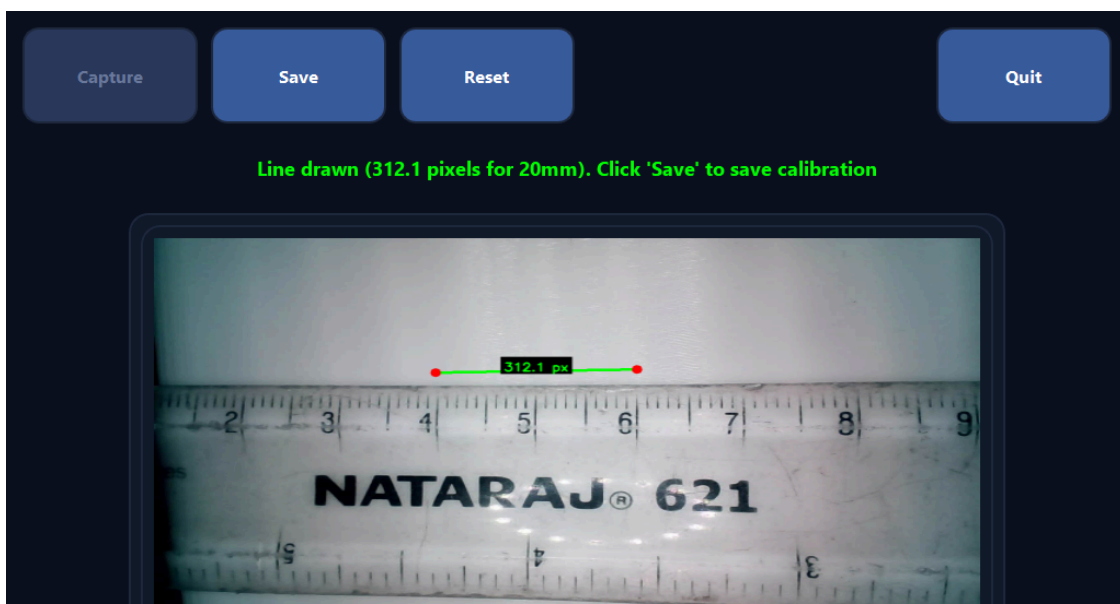
Why Calibrate? Calibration converts pixel measurements to real millimeters, ensuring accurate distance measurements.

How to Calibrate:

1. Click the **"Calibration"** button on the main screen
2. A new window will open showing the camera feed



3. Place a ruler with clear 20mm markings in the camera view
4. Click **"Capture"** to freeze the image
5. Click and drag to draw a line over exactly **20mm** on the ruler
 - Start at the 0mm mark
 - Drag to the 20mm mark
6. The system shows the pixel length in real-time



7. Click **"Save"** to store the calibration
8. Close the calibration window

Tips:

- ✓ *Ensure good lighting*
- ✓ *Keep the ruler flat and perpendicular to the camera*
- ✓ *Draw the line carefully for accuracy*
- ✓ *You'll see: "Calibration saved successfully!"*

Result: A file called `calibration.json` is created with your calibration data.

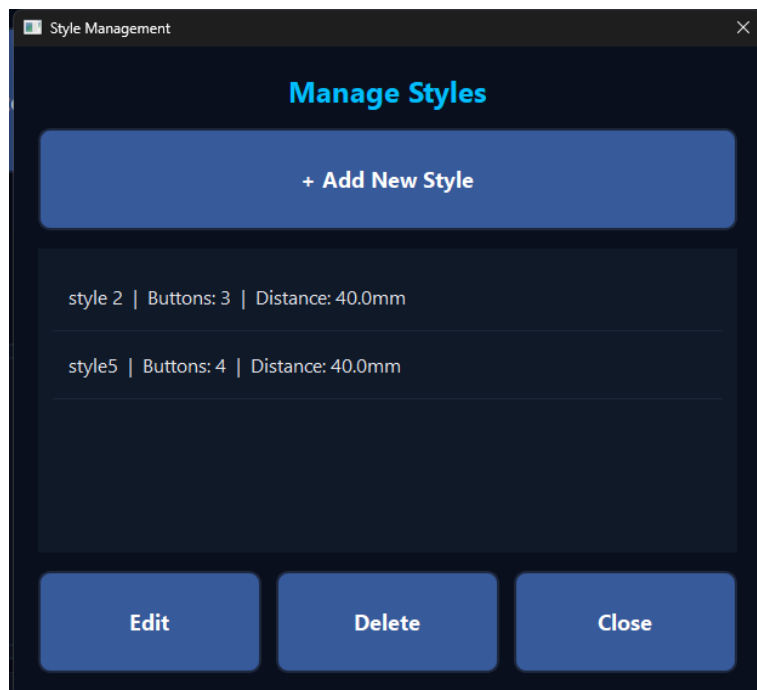
Creating Style Profiles

What is a Style Profile?

A style profile defines the expected characteristics for a specific garment type.

Step 1: Open Settings

1. Click the ⚙️ (**Settings**) button on the main screen
2. The "Style Management" window opens



Step 2: Add New Style

1. Click **" + Add New Style."**
2. Enter the following information:

Style Name (Full Keyboard)

- Click the  keyboard button
- Type a unique name (e.g., "Style 01"...)
- Click **"OK"**

Button Count (Numeric Keyboard)

- Click the  keyboard button
- Enter the expected number of buttons (e.g., 3, 5, 7)
- Click **"OK"**

Button Distance (Decimal Keyboard)

- Click the  keyboard button
 - Enter the maximum expected distance between buttons in millimeters
 - Example: 20 . 5 means buttons should be within 20.5mm of each other
 - Click **"OK"**
3. Click **"Save"** to store the style profile

Step 3: Managing Styles

Edit Existing Style:

1. Select the style from the list
2. Click **"Edit"**
3. Modify the values
4. Click **"Save"**

Delete Style:

1. Select the style from the list
2. Click **"Delete"**
3. Confirm deletion

Close Settings:

- Click **"Close"** to return to the main screen

Running Detection

Step 1: Select Style Profile

1. Look at the **Style dropdown** menu (below the main buttons)
2. Click the dropdown and select your desired style
3. The status bar shows: "✓ Style Selected: [Name] | Buttons: [Count] | Distance: [Value]mm."

 **Important:** You MUST select a style before starting detection!

Step 2: Start Detection

1. Click "**Start**"
2. The status indicator turns **GREEN** and shows "Status: Running."
3. Both camera panels activate:
 - **Left Panel:** Live camera stream
 - **Right Panel:** Detection results (initially empty)

Step 3: Monitoring

What You'll See:

Camera Stream (Left Panel):

- Shows "MONITORING" when waiting for motion
- Shows "WAITING" with a countdown when motion is detected
- Shows "Captures: X" count in top-left corner

Detection Result (Right Panel):

- Shows an annotated image after each capture
- Green boxes around detected buttons
- Red boxes around YOLO defects
- Black line showing maximum distance between buttons
- Distance measurement in millimeters
- Red banner at bottom if defects are found

Step 4: Automatic Operation

Motion Detection Sequence:

1. System monitors for movement
2. When motion is detected, "WAITING" appears with a 0.5s countdown
3. The system captures and analyzes images
4. Results displayed on the right panel

5. Returns to monitoring for the next garment

Step 5: Stop Detection

1. Click **"Stop"** when finished
2. Status indicator turns **RED** and shows "Status: Stopped"
3. Camera feed stops

Understanding Results

Detection Display

Button Annotations:

- **Green boxes** = Detected buttons (labeled "button 1", "button 2", etc.)
- **Red circles** = Button centers
- **Black line** = Maximum distance between buttons
- **Distance text** = Actual measurement in millimeters

Defect Annotations:

- **Red boxes with labels:**
 - "BEND PROUND 0.85" = Physical bend/round defect (confidence 85%)
 - "MACHINE 0.92" = Machine damage defect (confidence 92%)

Alert Banner

Red Banner at Bottom: Shows when defects are detected. Examples:

- BUTTON COUNT MISMATCH: Expected 3, Found 2
- DISTANCE OUT OF RANGE: 25.3mm (Expected: 20.0mm $\pm$ 3mm)
- BEND PROUND DEFECT (conf: 0.85)
- MACHINE DEFECT (conf: 0.92)
- COLOR DEFECT $\times$ 1

No Banner = All checks passed 

IMPORTANT SAFETY INFORMATION

READ THIS BEFORE OPERATING THE SYSTEM

Improper shutdown can cause:

- Data corruption
- File system damage

Power ON Procedure

Step 1: Visual Inspection

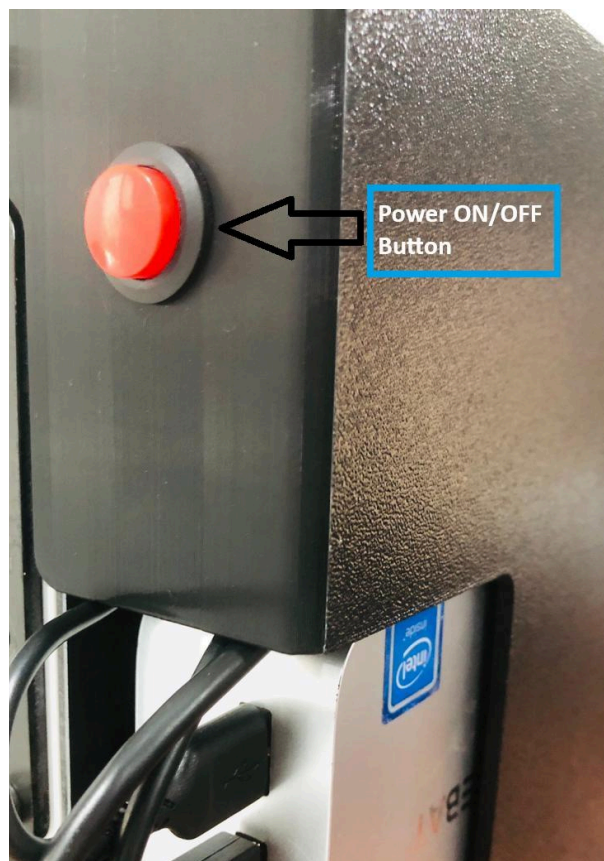
Check that all cables are properly connected:

- Power cable securely plugged in
- Camera USB cable connected
- Touchscreen display cable connected
- Serial cable (COM3) connected

Step 2: Power Up Sequence

Plug in the power cable to the electrical outlet

- Press the power button
- Wait for the system to boot (30-60 seconds)



- You'll see boot messages on screen
- LEDs will illuminate

Power OFF Procedure

 **CRITICAL: NEVER UNPLUG WITHOUT PROPER SHUTDOWN!**

Step 1: Stop All Operations

If detection is running:

- Click the "Stop Detection" button
- Wait for the status to show "Status: Stopped."
- Verify camera feeds have stopped

Step 2: Proper System Shutdown

Initiate system shutdown:

- Press Power Button → Shutdown
- The system will close all programs
- The display will show the "Shutting down..." message

Step 3: Wait for All Lights to Turn OFF

WAIT 30-60 SECONDS after shutdown command

- Monitor the indicator lights: Should turn OFF
- Display: Should go completely dark

Step 4: Safe to Unplug

Only after ALL lights are OFF and the system is silent:

- Now it's safe to unplug the power cable

IMPORTANT SAFETY RULES

NEVER:

- Unplug the power cable while the system is running
- Unplug while the lights are still ON
- Disconnect cables while powered on

ALWAYS:

- Use the proper shutdown procedure
- Wait for all lights to turn OFF
- Give the system 30-60 seconds after shutdown